

Learning Dynamics and Market Outcomes in Algorithmic Bertrand Competition

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Abstract

We study the pricing behaviour of reinforcement-learning agents in repeated Bertrand competition with *homogeneous* goods — the environment in which classical theory predicts the sharpest competitive outcome. Using ε -greedy Q-learning with state given by rivals’ previous prices, and a stateless mean-based (bandit) benchmark, we run 100 independent simulations per configuration and classify outcomes with a Shannon-entropy scheme into competitive, collusive, and chaotic regimes. Three findings stand out. First, supra-competitive collusion does emerge even in the homogeneous-goods case — in 56% of baseline duopoly runs, at roughly half the monopoly markup ($\Delta \approx 0.43$) — but it is *conditional*, not universal: it requires a tractable action space, annealed exploration, and a sufficiently long horizon, and it collapses into non-convergent (chaotic) pricing otherwise. Second, and contrary to the intuition that algorithmic sophistication drives collusion, the *simpler* mean-based learner collude more reliably (89% of runs) than Q-learning (56%); the bootstrapped Q-learner is more prone to non-convergence. Third, the intensity of collusion is governed by the discount factor in a manner consistent with the folk theorem: the collusion index rises monotonically with patience (Δ from 0.23 at $\gamma = 0.2$ to 0.45 at $\gamma = 0.95$). Collusion weakens monotonically as the price grid fines, as exploration persists, as the horizon shortens, and as a third firm is added. The policy implication is sharpened rather than softened: tacit algorithmic collusion does not require sophisticated algorithms, but in the hardest competitive environment it is fragile and condition-dependent.

Keywords: algorithmic collusion, Bertrand competition, reinforcement learning, Q-learning, tacit collusion, computational economics, competition policy.

JEL: D43, L13, L41, C63.

1 Introduction

The Bertrand model yields one of the sharpest predictions in economic theory: with a homogeneous good and identical constant marginal costs, the unique Nash equilibrium has all firms pricing at marginal cost and earning zero profit, so that even a duopoly reproduces the perfectly competitive outcome [Tirole, 1988]. As firms increasingly delegate pricing to autonomous algorithms, a first-order question for competition authorities is whether such algorithms can learn to sustain supra-competitive prices without being instructed to do so [Harrington, 2018, Ezrachi and Stucke, 2016].

Calvano et al. [2020] showed, in a *differentiated*-product setting, that Q-learning agents converge to supra-competitive prices and develop reward–punishment strategies. We ask the harder question:

does this occur in the *homogeneous*-goods environment, where the competitive force is strongest and the Bertrand paradox bites? Picking the hardest case is deliberate: collusion “even here” would be a stronger and more alarming result; the failure of collusion here would delimit the phenomenon.

We implement faithful ε -greedy Q-learning — with a temporal-difference update, a discount factor, and state given by rivals’ previous prices — alongside a stateless mean-based (multi-armed bandit) benchmark, and study 100 independent runs per configuration. We classify each run into one of three regimes using a Shannon-entropy measure of price dispersion, and we summarise collusion with a normalised index $\Delta \in [0, 1]$ running from Bertrand–Nash (0) to symmetric joint monopoly (1).

Our findings revise, rather than echo, the differentiated-goods literature:

1. **Conditional collusion.** Collusion emerges even with homogeneous goods — 56% of baseline duopoly runs, $\Delta \approx 0.43$ — but it is not the universal outcome. The remaining runs are *chaotic* (non-convergent). Collusion requires a tractable price grid, annealed exploration, and a long horizon; outside those conditions the dominant outcome is chaos.
2. **Sophistication is not the driver.** The simple mean-based bandit colludes *more* often (89%) than Q-learning (56%). The temporal-difference bootstrap that defines Q-learning makes convergence harder here, not easier. Collusion is therefore a generic property of payoff-maximising learning, not an artefact of algorithmic sophistication.
3. **Patience governs intensity.** The collusion index rises monotonically with the discount factor γ ($\Delta : 0.23 \rightarrow 0.45$ as $\gamma : 0.2 \rightarrow 0.95$), the comparative static predicted by the folk theorem.
4. **Fragility.** Collusion falls monotonically as the grid fines (57% $\rightarrow$ 3%), as exploration persists (57% $\rightarrow$ 0%), as the horizon shortens (57% $\rightarrow$ 10%), and as a third firm is added (57% $\rightarrow$ 0%).

The policy reading is that tacit algorithmic collusion is real and does not require clever algorithms — but in the homogeneous-goods limit it lives on a knife-edge of conditions, which both tempers the most alarmist claims and points to concrete, observable levers (exploration floors, market structure) for regulation.

2 Related Literature

Our work sits at the intersection of classical Bertrand theory, the algorithmic-collusion literature, and learning in games. The Bertrand paradox [Bertrand, 1883, Tirole, 1988] is relaxed classically by product differentiation [Hotelling, 1929], capacity constraints [Edgeworth, 1925], and repeated interaction [Ivaldi et al., 2003]; we study adaptive algorithmic learning as a further channel. Calvano et al. [2020] is the closest antecedent: Q-learning collusion with differentiated products and reward–punishment strategies. Klein [2021] studies sequential pricing; Brown and MacKay [2023] analyse competition between pricing algorithms; Assad et al. [2024] provide empirical evidence from retail gasoline. On the methodological side, Waltman and Kaymak [2008] document that multi-agent Q-learning may cycle rather than converge — precisely the chaotic regime we find to be dominant in the homogeneous case — and Watkins and Dayan [1992] introduce the algorithm we use. Relative to this literature our contribution is to (i) push the analysis to the homogeneous-goods limit, (ii) provide an entropy-based regime classifier, and (iii) show that the *simpler* learner colludes more, reframing collusion as independent of algorithmic sophistication.

3 Model

We consider a repeated Bertrand pricing game with $n \in \{2, 3\}$ firms producing a homogeneous good. Each firm i chooses a price p_i from a finite, evenly spaced grid

$$\mathcal{P} = \{p_{\min}, p_{\min} + \Delta, \dots, p_{\max}\}, \quad \Delta = \frac{p_{\max} - p_{\min}}{K - 1}, \quad K = |\mathcal{P}|,$$

with $p_{\min} = 0$, $p_{\max} = 10$, and a baseline of $K = 7$ price points (Section 9 varies K). All firms share constant marginal cost $c = 1$. Demand is inelastic and normalised to one unit, allocated by the Bertrand rule

$$q_i(\mathbf{p}) = \begin{cases} 1/|\{j : p_j = \min_k p_k\}| & \text{if } p_i = \min_k p_k, \\ 0 & \text{otherwise,} \end{cases} \quad \pi_i(\mathbf{p}) = (p_i - c) q_i(\mathbf{p}).$$

The stage-game Nash equilibrium is $p_i = c$ for all i , with zero profit. Firms interact for T periods (baseline $T = 30,000$; Section 9 varies T). We summarise the degree of collusion by the index

$$\Delta = \frac{\bar{\pi} - \pi^N}{\pi^M - \pi^N}, \quad \pi^N = 0, \quad \pi^M = \frac{p_{\max} - c}{n},$$

where $\bar{\pi}$ is the average per-firm per-period profit over the final $T_0 = 2,000$ periods; $\Delta = 0$ is Bertrand–Nash and $\Delta = 1$ is symmetric joint monopoly.

4 Learning Algorithms

Q-learning. Each firm runs ε -greedy Q-learning [Watkins and Dayan, 1992]. The state s_t is the vector of rivals' prices in the previous period; the action is the firm's own price. With probability $1 - \varepsilon$ the firm plays $\arg \max_a Q^i(s_t, a)$, and with probability ε it explores uniformly. After observing profit π_t^i and the new state s_{t+1} , it applies the temporal-difference update

$$Q^i(s_t, a_t^i) \leftarrow Q^i(s_t, a_t^i) + \alpha \left[\pi_t^i + \gamma \max_{a'} Q^i(s_{t+1}, a') - Q^i(s_t, a_t^i) \right], \quad (1)$$

with learning rate $\alpha = 0.10$ and discount factor $\gamma = 0.95$ (baseline). Exploration is annealed, $\varepsilon_t = \varepsilon_0 / (1 + \delta t)$ with $\varepsilon_0 = 0.10$ and $\delta = 3 \times 10^{-4}$; Section 9 compares constant and annealed schedules. Q-tables are initialised at (numerically) zero. Unlike a bandit, (1) is forward-looking (through $\gamma \max_{a'} Q$) and state-dependent, in principle enabling reactive strategies.

Mean-based benchmark. As a control we use stateless mean-based learning: each agent keeps the running average payoff of each price, $M_t^i(p)$, and plays ε -greedily with respect to it. This is a multi-armed bandit: it has no discounting, no state, and no temporal-difference bootstrap. Comparing the two isolates the role of Q-learning's added machinery.

5 Behavioural Regimes and Entropy Classification

Three qualitatively distinct long-run behaviours arise (Figure 1): a *competitive* regime with prices near marginal cost and $\Delta \approx 0$; a *collusive* regime with both firms converging to a common supra-competitive price; and a *chaotic* regime of persistent oscillation without convergence.

Figure 1: Representative price trajectories (dashed = marginal cost)

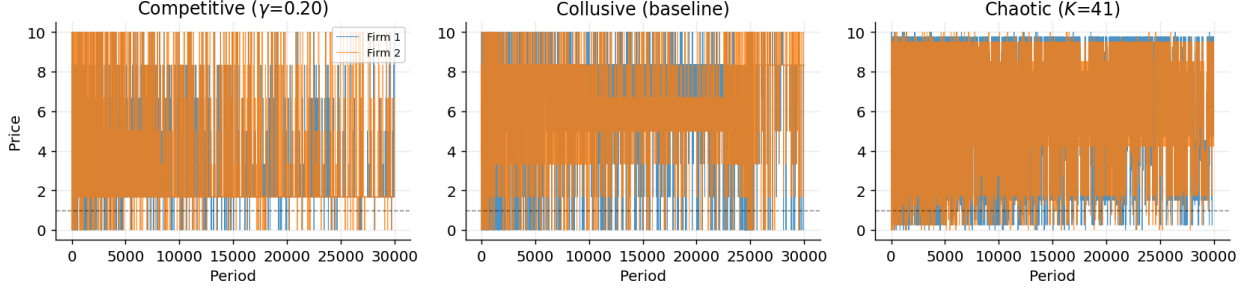


Figure 1: Representative price trajectories. Left: competitive ($\gamma = 0.20$). Centre: collusive (baseline). Right: chaotic (fine grid, $K = 41$). Dashed line is marginal cost $c = 1$.

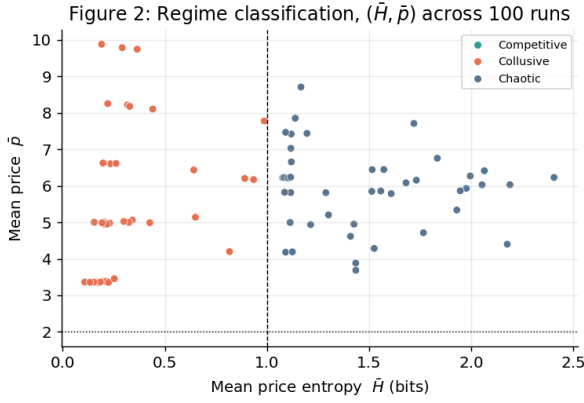


Figure 2: Regime classification of 100 baseline runs in $(\bar{H}, \bar{p})$.

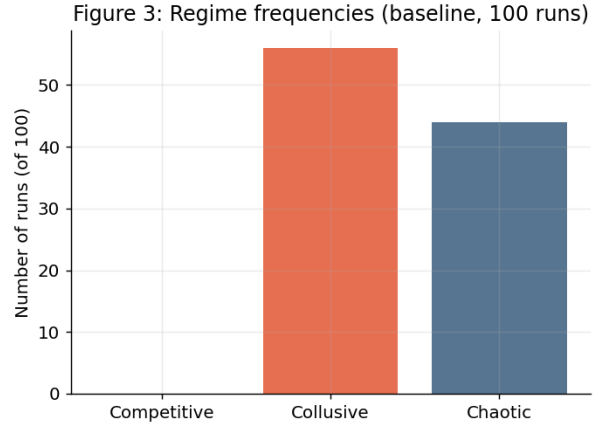


Figure 3: Baseline regime frequencies (of 100 runs).

To classify runs automatically we compute, over the final T_0 periods, the empirical price distribution $f^i(p)$ of each agent and its Shannon entropy $H^i = -\sum_p f^i(p) \log_2 f^i(p)$, averaged to $\bar{H} = \frac{1}{n} \sum_i H^i$. Writing $\bar{p}$ for the mean price over the window, the regime is

$$\text{Regime} = \begin{cases} \text{Competitive} & \bar{H} < H^* \text{ and } \bar{p} < p^*, \\ \text{Collusive} & \bar{H} < H^* \text{ and } \bar{p} \geq p^*, \\ \text{Chaotic} & \bar{H} \geq H^*, \end{cases} \quad H^* = 1.0, p^* = 2.0.$$

Entropy distinguishes converged play (low $\bar{H}$) from non-convergent play (high $\bar{H}$); mean price then separates competitive from collusive convergence. Figure 2 shows the 100 baseline runs in the $(\bar{H}, \bar{p})$ plane, with the two clusters — a converged collusive mass and a high-entropy chaotic mass — clearly separated by H^* .

6 Baseline Results

Under the baseline duopoly configuration, 56% (± 5.0) of runs are collusive and 44% chaotic; competitive convergence essentially never occurs (Figure 3). The mean collusion index is $\Delta = 0.43$, i.e. converged runs sit roughly half-way between marginal-cost pricing and joint monopoly. The

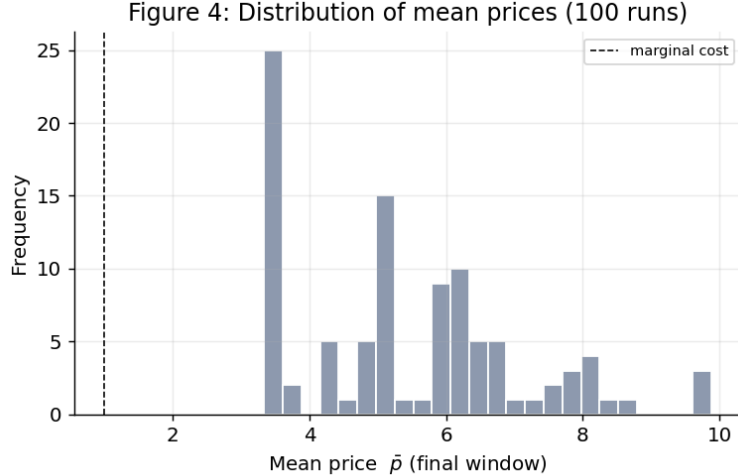


Figure 4: Distribution of mean prices across 100 baseline runs (dashed line: marginal cost).

distribution of mean prices (Figure 4) is bimodal: a mass of converged collusive runs at high prices and a spread of chaotic runs at intermediate prices, with almost no mass at marginal cost. Thus collusion is the single most common *converged* outcome, but non-convergence is nearly as likely: in the homogeneous-goods limit, learning to collude and failing to settle are comparably probable.

7 Q-learning versus Mean-Based Learning

Comparing the two algorithms under identical conditions yields our most counterintuitive result (Figure 5): the stateless mean-based bandit colludes in 89% of runs, against only 56% for Q-learning. The added sophistication of Q-learning — the temporal-difference bootstrap and the enlarged state space — makes convergence *harder*, not easier, so a larger share of Q-learning runs end chaotic. Mean-based agents, by contrast, settle quickly onto the price with the highest historical average payoff, which under symmetric play is supra-competitive, and stay there.

This reverses the natural hypothesis that collusion is a product of algorithmic sophistication. In the homogeneous-goods environment, collusion is if anything the *default* of naive payoff-averaging, and the more elaborate learner is more easily destabilised. For competition policy this is the more uncomfortable reading: supra-competitive coordination does not require firms to deploy advanced reinforcement learning; a simple bandit suffices.

8 Patience and the Intensity of Collusion

The discount factor γ governs how heavily an agent weighs future payoffs, and hence its willingness to forgo a one-period undercut to preserve an ongoing arrangement. Consistent with the folk theorem [Tirole, 1988], the intensity of collusion rises monotonically with patience: the mean collusion index climbs from $\Delta = 0.23$ at $\gamma = 0.20$ to $\Delta = 0.45$ at $\gamma = 0.95$ (Figure 6). Impatient agents ($\gamma = 0.20$) still coordinate frequently but only weakly, and are the only setting in which the competitive regime appears with non-trivial probability; patient agents collude more intensely but, because they pursue higher and more fragile prices, also fall into chaos more often. The Q-

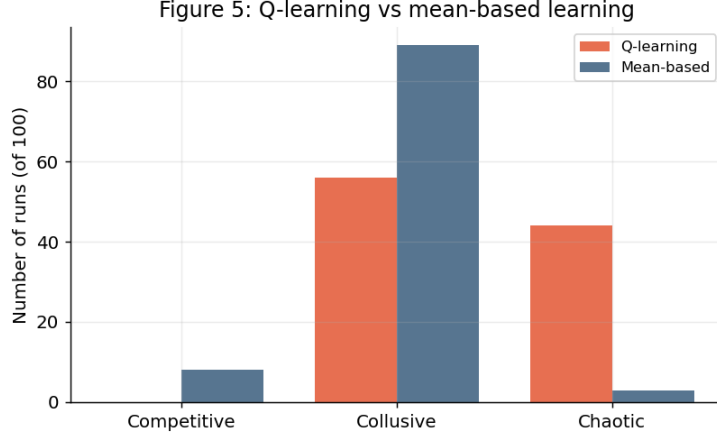


Figure 5: Regime frequencies under Q-learning versus mean-based learning (100 runs each).

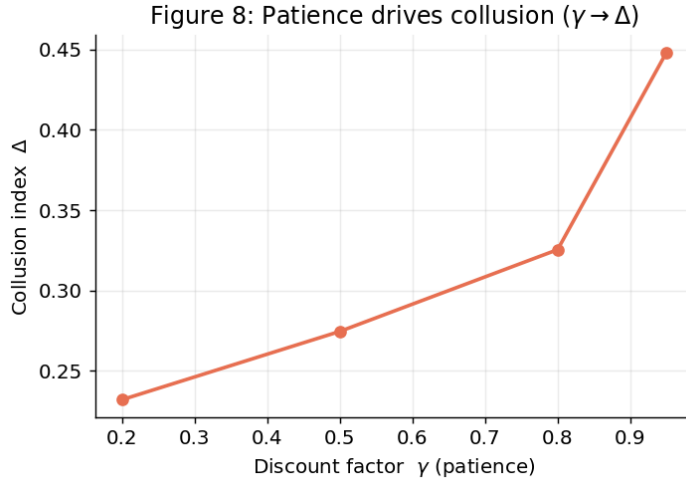


Figure 6: Collusion index Δ as a function of the discount factor γ (patience).

learner thus discovers folk-theorem-like behaviour through adaptive search, with γ playing the role of patience — without any knowledge of repeated-game theory.

9 Robustness and the Fragility of Collusion

Table 1 reports the share of collusive runs and the mean collusion index as we vary, one at a time, the discount factor, exploration schedule, grid granularity, horizon, and number of firms (30 runs per cell). Four comparative statics emerge sharply, all in the direction theory predicts.

Grid granularity. Collusion is highly sensitive to the size of the action space: the collusive share falls from 57% at $K = 7$ to 3% at $K = 41$. A finer grid enlarges the state–action space faster than agents can learn it within the horizon, so the dominant outcome becomes chaotic. This is the central reason a naive “fine grid, short horizon” specification finds no collusion at all.

Table 1: Sensitivity analysis. Collusive share (% of 30 runs) and mean collusion index Δ . Baseline: $n=2$, $K=7$, $\gamma=0.95$, annealed ε , $T=30,000$.

Parameter	Value	Collusive (%)	Δ
Discount γ	0.20	83	0.23
	0.50	97	0.27
	0.80	93	0.33
	0.95 (base)	57	0.45
Exploration ε	annealed (base)	57	0.45
	constant 0.10	13	0.32
	constant 0.30	0	0.23
Grid K	7 (base)	57	0.45
	11	10	0.33
	21	3	0.33
	41	3	0.30
Horizon T	5,000	10	0.33
	15,000	30	0.39
	30,000 (base)	57	0.45
	60,000	77	0.49
Firms n	2 (base)	57	0.45
	3	0	0.23

Exploration. Annealing exploration is necessary for stable collusion. With constant $\varepsilon = 0.10$ the collusive share is only 13%, and with $\varepsilon = 0.30$ it is zero: persistent random perturbation repeatedly destabilises any incipient arrangement. The natural tendency of an efficient algorithm to reduce exploration as it grows confident is therefore precisely the feature that enables collusion — and a mandated exploration floor is a candidate pro-competitive intervention.

Horizon. Collusion rises monotonically with the horizon (10% at $T = 5,000$ to 77% at $T = 60,000$), confirming that short-horizon specifications understate collusion simply because learning has not converged.

Market structure. Adding a third firm collapses collusion entirely ($57\% \rightarrow 0\%$, all runs chaotic): the joint state space grows combinatorially and the incentive to undercut two rivals rises, making coordination far more fragile. Figure 7 illustrates a representative three-firm run.

10 Discussion

Three implications follow. First, the Bertrand paradox is not robust to adaptive learning even with homogeneous goods: payoff-maximising algorithms reach supra-competitive prices in a majority of converged runs, with no communication or instruction to collude. Second, because the *simpler* learner colludes more, the phenomenon is not a quirk of advanced reinforcement learning; it is a generic tendency of independent payoff optimisation, which broadens rather than narrows the policy concern. Third, however, in the homogeneous-goods limit collusion is fragile and condition-dependent — it requires a tractable action space, annealed exploration, a long horizon, and few

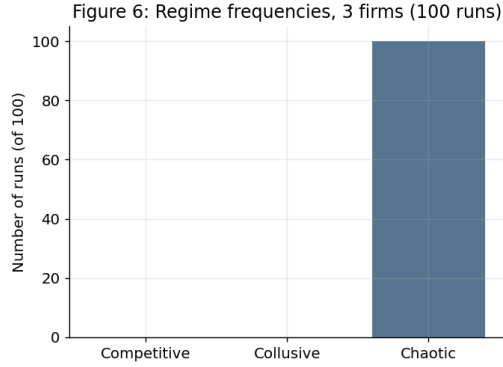


Figure 7: Regime frequencies with three firms (100 runs).

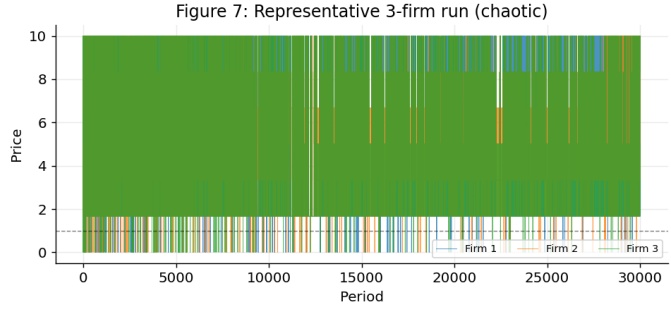


Figure 8: Representative three-firm price trajectory (chaotic).

firms — so the differentiated-goods alarm of Calvano et al. [2020] should be read as an upper bound on how easily algorithmic collusion arises.

For enforcement, the conduct-based framework that requires evidence of agreement is ill-suited to a setting where coordination emerges from independent learning with no communication [Harrington, 2018, Ezrahi and Stucke, 2016]. Our results point to observable, structural levers: exploration floors (which we find directly prevent convergence to collusion), market-structure thresholds (collusion vanishes at three firms), and the coarseness of admissible price grids.

Limitations. The analysis is computational and offers no convergence proofs; multi-agent Q-learning is known to cycle rather than converge in general [Waltman and Kaymak, 2008], which our chaotic regime reflects. We restrict attention to homogeneous goods, inelastic demand, tabular learners, and memory-one state; richer demand, deep reinforcement learning, and longer memory are natural extensions. The collusive share depends on the entropy threshold H^* and price threshold p^* ; we report the comparative statics, which are robust to moderate threshold changes, rather than treating any single frequency as exact.

11 Conclusion

Implementing faithful Q-learning and a mean-based benchmark in homogeneous-good Bertrand competition, we find that supra-competitive collusion does emerge — in a majority of converged duopoly runs, at about half the monopoly markup — but that it is conditional rather than universal, that it does not require algorithmic sophistication (a simple bandit colludes more), that its intensity tracks patience as the folk theorem predicts, and that it collapses under fine price grids, persistent exploration, short horizons, or additional firms. Tacit algorithmic collusion is thus both more accessible (it needs no clever algorithm) and more fragile (it lives on a knife-edge of conditions) than a reading of the differentiated-goods literature alone would suggest.

Reproducibility. All code, data, and figure-generating scripts are available at <https://github.com/shubhL-research/algorithmic-bertrand-rl>; every result is produced from fixed random seeds by a single script.

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